



Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

LESSON PLAN

For

Cisco Certified Network Associate (CCNA)

TGS Ref No: TGS-2023037854

Conducted by

Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

Version v7.0

DOCUMENT VERSION CONTROL RECORD

Version Number	Effective Date of Release	Summary of Included Changes	Author
6.0	Legacy reference version	Original CCNA course material retained privately for coverage comparison.	Tertiary Infotech Academy
7.0	13 September 2026	Rebuilt for current CCNA v1.1,announced v2.0 transition,aligned activities and verified evidence.	Tertiary Infotech Academy

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Programme Structure and Outcomes

Course TGS-2023037854 ·5 days / 40 hours. Training:19 lecture +19 practical = 38 hours.

Assessment:Written 1 hour + Practical 1 hour = 2 hours. Breaks are outside allocated contact time;daily tables use session minutes rather than wall-clock times. Trainer flexes discussion pace across substantive evidence slides while preserving domain coverage and activity duration.

Course outcome	TSC alignment
Requirements, component dependencies, addressing/capacity evidence	K1/A1-A2
Device/server configuration and implementation techniques	K2-K4/A3
Management access and network security control	K5/A4
Tool selection, tracking and stability reporting	K3/K6/A5-A6

Day 1

Type	Session	Minutes	Slides	Method / evidence
Lecture	Network Fundamentals	300	Slides 16-161	Technical exhibit discussion, worked paths and diagnostic evidence
Practical	Activity 1 — Addressing and endpoint paths	90	Slide 701	Guided implementation, fixture check, hardware/simulator evidence and checklist
Practical	Activity 2 — Physical and VLAN evidence	90	Slide 702	Guided implementation, fixture check, hardware/simulator evidence and checklist

Allocated contact time: 480 minutes. Attendance/admin checks are integrated;breaks add time outside the allocation.

Day 2

Type	Session	Minutes	Slides	Method / evidence
Lecture	Network Access	190	Slides 163-283	Technical exhibit discussion, worked paths and diagnostic evidence
Practical	Activity 3 — Trunks and inter-VLAN routing	100	Slide 703	Guided implementation, fixture check, hardware/simulator evidence and checklist
Practical	Activity 4 — EtherChannel and STP resilience	100	Slide 704	Guided implementation, fixture check, hardware/simulator evidence and checklist
Practical	Activity 5 — Wireless WLAN and policy mapping	90	Slide 705	Guided implementation, fixture check, hardware/simulator evidence and checklist

Allocated contact time: 480 minutes. Attendance/admin checks are integrated;breaks add time outside the allocation.

Day 3

Type	Session	Minutes	Slides	Method / evidence
Lecture	IP Connectivity	260	Slides 284-414	Technical exhibit discussion, worked paths and diagnostic evidence
Practical	Activity 6 — Static routing and IPv6	100	Slide 706	Guided implementation, fixture check, hardware/simulator evidence and checklist
Practical	Activity 7 — OSPF and gateway failover	120	Slide 707	Guided implementation, fixture check, hardware/simulator evidence and checklist

Allocated contact time: 480 minutes. Attendance/admin checks are integrated; breaks add time outside the allocation.

Day 4

Type	Session	Minutes	Slides	Method / evidence
Lecture	IP Services	80	Slides 415-480	Technical exhibit discussion, worked paths and diagnostic evidence
Lecture	Security Fundamentals	100	Slides 481-571	Technical exhibit discussion, worked paths and diagnostic evidence
Practical	Activity 8 — NAT DHCP DNS and time	100	Slide 708	Guided implementation, fixture check, hardware/simulator evidence and checklist
Practical	Activity 9 — ACL and secure management	110	Slide 709	Guided implementation, fixture check, hardware/simulator evidence and checklist
Practical	Activity 10 — Layer 2 and wireless security	90	Slide 710	Guided implementation, fixture check, hardware/simulator evidence and checklist

Allocated contact time: 480 minutes. Attendance/admin checks are integrated; breaks add time outside the allocation.

Day 5

Type	Session	Minutes	Slides	Method / evidence
Lecture	Automation and Programmability	110	Slides 572-622	Technical exhibit discussion, worked paths and diagnostic evidence
Lecture	Announced v2.0 transition	100	Slides 625-700	Technical exhibit discussion, worked paths and diagnostic

				evidence
Practical	Activity 11 — REST JSON and configuration automation	80	Slide 711	Guided implementation, fixture check, hardware/simulator evidence and checklist
Practical	Activity 12 — Integrated incident and change verification	70	Slide 712	Guided implementation, fixture check, hardware/simulator evidence and checklist
Assessment	Written Assessment	60	—	Approved candidate paper; knowledge K1-K6.
Assessment	Practical Performance	60	—	Approved performance tasks; abilities A1-A6.

Allocated contact time: 480 minutes. Attendance/admin checks are integrated; breaks add time outside the allocation.

Topic-to-slide Delivery Map

Network Fundamentals - Slides 16-161

Current v1.1 weighting 20%. Teach boundaries, runtime, exact contract, failure/control and verification for each topic. The deck contains five-slide technical units; trainer uses full exhibits and questions to test interpretation.

Technical unit	Slides
Frame and packet boundaries	17-21
Router and switch forwarding	22-26
Firewall and IPS control points	27-31
Wireless controller boundaries	32-36
Endpoint and server dependencies	37-41
PoE power budgets	42-46
Two-tier and three-tier paths	47-51
Spine-leaf forwarding	52-56
WAN and SOHO boundaries	57-61
Cloud and on-premises ownership	62-66
Copper and optical link contracts	67-71
Duplex and interface errors	72-76
TCP session state	77-81
UDP delivery and application recovery	82-86
IPv4 subnet boundaries	87-91
VLSM address allocation	92-96
Private and public addressing	97-101
IPv6 address types	102-106
IPv6 neighbor and SLAAC state	107-111
Wireless channel and virtual boundaries	112-116
IPv6 interface and anycast contracts	117-121
VRF routing isolation	122-126
MAC aging and unknown-unicast scope	127-131
Modified EUI-64 address construction	132-136
Hypervisor VM and container isolation	137-141
RF link-budget and interference evidence	142-146
Cross-platform client diagnostics	147-151
Five equal subnets from a /24	152-156
Single-mode versus multimode optical paths	157-161

Network Access •Slides163-283

Current v1.1 weighting 20%. Teach boundaries, runtime, exact contract, failure/control and verification for each topic. The deck contains five-slide technical units;trainer uses full exhibits and questions to test interpretation.

Technical unit	Slides
Access VLAN membership	164-168
Voice and data separation	169-173
802.1Q trunk tagging	174-178
Native VLAN consistency	179-183
Trunk negotiation boundaries	184-188
Router-on-a-stick paths	189-193
Multilayer SVI routing	194-198
CDP neighbor evidence	199-203
LLDP interoperable discovery	204-208
LACP channel negotiation	209-213
EtherChannel load distribution	214-218
STP root bridge election	219-223
STP root and designated ports	224-228
Rapid PVST convergence	229-233
PortFast and BPDU guard	234-238
Root guard boundary	239-243
Loop guard and BPDU filter	244-248
AP modes and WLAN transport	249-253
WLAN GUI policy evidence	254-258
Wireless management reachability	259-263
Layer 3 LACP port-channel	264-268
Default and normal-range VLAN contracts	269-273
AP and WLC physical attachments	274-278
WLAN QoS and advanced-policy evidence	279-283

IP Connectivity •Slides284-414

Current v1.1 weighting 25%. Teach boundaries, runtime, exact contract, failure/control and verification for each topic. The deck contains five-slide technical units;trainer uses full exhibits and questions to test interpretation.

Technical unit	Slides
Route-table field interpretation	285-289
Longest-prefix matching	290-294
Administrative distance selection	295-299
Metric comparison boundaries	300-304
Connected and local routes	305-309
Static next-hop recursion	310-314
Fully specified static routes	315-319
Default IPv4 routing	320-324
Host-specific static routes	325-329
Floating static backup	330-334
IPv6 static-route scope	335-339
IPv6 default and host routes	340-344
OSPF process and area boundaries	345-349
OSPF interface activation	350-354
OSPF router ID selection	355-359
OSPF Hello compatibility	360-364
OSPF neighbor state exchange	365-369
OSPF point-to-point operation	370-374
OSPF DR and BDR election	375-379
OSPF cost calculation	380-384
OSPF LSA and SPF boundaries	385-389
OSPF failure diagnosis	390-394

Equal-cost multipath forwarding	395-399
FHRP virtual-gateway ownership	400-404
End-to-end routed fault isolation	405-409
IPv6 floating static failover	410-414

IP Services •Slides415-480

Current v1.1 weighting 10%. Teach boundaries, runtime, exact contract, failure/control and verification for each topic. The deck contains five-slide technical units;trainer uses full exhibits and questions to test interpretation.

Technical unit	Slides
Static NAT mapping	416-420
Dynamic NAT pool allocation	421-425
PAT port disambiguation	426-430
NTP synchronization hierarchy	431-435
DHCP DORA allocation	436-440
DHCP relay and client roles	441-445
DNS record and cache evidence	446-450
SNMP manager and agent state	451-455
Syslog severity and facility	456-460
QoS classification and management transfer	461-465
NTP server and client association	466-470
IOS DHCP client interface state	471-475
QoS congestion decisions	476-480

Security Fundamentals •Slides481-571

Current v1.1 weighting 15%. Teach boundaries, runtime, exact contract, failure/control and verification for each topic. The deck contains five-slide technical units;trainer uses full exhibits and questions to test interpretation.

Technical unit	Slides
Threat-vulnerability-exploit chain	482-486
Security program evidence	487-491
Local user and enable secrets	492-496
SSH management prerequisites	497-501
Password and MFA policy	502-506
AAA responsibility split	507-511
RADIUS and TACACS+ boundaries	512-516
Site-to-site IPsec trust	517-521
Remote-access VPN roles	522-526
Standard ACL source matching	527-531
Extended ACL tuple control	532-536
ACL wildcard mathematics	537-541
Port-security admission and violations	542-546
DHCP snooping and ARP inspection	547-551
Wireless WPA and authentication	552-556
Physical access and identity controls	557-561
WPA lineage and WPA2 PSK WLAN evidence	562-566
IPsec tunnel and transport boundaries	567-571

Automation and Programmability •Slides572-622

Current v1.1 weighting 10%. Teach boundaries, runtime, exact contract, failure/control and verification for each topic. The deck contains five-slide technical units;trainer uses full exhibits and questions to test interpretation.

Technical unit	Slides
Traditional and controller operation	573-577
Control and data plane separation	578-582

Overlay underlay and fabric roles	583-587
REST resource and HTTP verbs	588-592
REST authentication and HTTPS	593-597
JSON typed data contracts	598-602
Ansible desired configuration	603-607
Terraform plan and state	608-612
Generative and predictive AI evidence	613-617
Automation change verification	618-622

Activity Delivery and Acceptance

Activity1 — Addressing and endpoint paths •Slide701

Allocate non-overlapping VLSM subnets and explain remote packet versus next-hop frame addressing.

Offline checker validates synthetic fixtures only. Trainer separately observes simulator/hardware operation, configuration evidence, diagnosis and safe-change discipline. Use the independent folder instructions and checklist to record acceptance.

Activity2 — Physical and VLAN evidence •Slide702

Use interface counter deltas and endpoint VLAN membership to separate physical faults from segmentation errors.

Offline checker validates synthetic fixtures only. Trainer separately observes simulator/hardware operation, configuration evidence, diagnosis and safe-change discipline. Use the independent folder instructions and checklist to record acceptance.

Activity3 — Trunks and inter-VLAN routing •Slide703

Diagnose a missing allowed VLAN and native VLAN mismatch, then verify SVI gateway mapping.

Offline checker validates synthetic fixtures only. Trainer separately observes simulator/hardware operation, configuration evidence, diagnosis and safe-change discipline. Use the independent folder instructions and checklist to record acceptance.

Activity4 — EtherChannel and STP resilience •Slide704

Check LACP negotiation and spanning-tree redundancy without confusing blocked paths with failed links.

Offline checker validates synthetic fixtures only. Trainer separately observes simulator/hardware operation, configuration evidence, diagnosis and safe-change discipline. Use the independent folder instructions and checklist to record acceptance.

Activity5 — Wireless WLAN and policy mapping •Slide705

Connect SSID, authentication, VLAN and client addressing while inspecting RF channel and WLC data-path choices.

Offline checker validates synthetic fixtures only. Trainer separately observes simulator/hardware operation, configuration evidence, diagnosis and safe-change discipline. Use the independent folder instructions and checklist to record acceptance.

Activity6 — Static routing and IPv6 ·Slide706

Apply longest-prefix match, a floating route and interface-scoped IPv6 link-local next hops.

Offline checker validates synthetic fixtures only. Trainer separately observes simulator/hardware operation, configuration evidence, diagnosis and safe-change discipline. Use the independent folder instructions and checklist to record acceptance.

Activity7 — OSPF and gateway failover ·Slide707

Diagnose OSPF adjacency, path costs and a virtual gateway; distinguish current v1.1 FHRP explanation from refresh practice.

Offline checker validates synthetic fixtures only. Trainer separately observes simulator/hardware operation, configuration evidence, diagnosis and safe-change discipline. Use the independent folder instructions and checklist to record acceptance.

Activity8 — NAT DHCP DNS and time ·Slide708

Correlate translated transport tuples, DHCP scope options, DNS records and synchronized logging.

Offline checker validates synthetic fixtures only. Trainer separately observes simulator/hardware operation, configuration evidence, diagnosis and safe-change discipline. Use the independent folder instructions and checklist to record acceptance.

Activity9 — ACL and secure management ·Slide709

Evaluate ordered ACL rules and secure VTY access, including AAA client/server roles and safe fallback.

Offline checker validates synthetic fixtures only. Trainer separately observes simulator/hardware operation, configuration evidence, diagnosis and safe-change discipline. Use the independent folder instructions and checklist to record acceptance.

Activity10 — Layer2 and wireless security ·Slide710

Map trusted infrastructure ports and untrusted edge ports to DHCP snooping, DAI, port security and wireless authentication.

Offline checker validates synthetic fixtures only. Trainer separately observes simulator/hardware operation, configuration evidence, diagnosis and safe-change discipline. Use the independent folder instructions and checklist to record acceptance.

Activity11 — REST JSON and configuration automation ·Slide711

Validate REST status and JSON types, produce an idempotent plan and select a bounded agentic AI prompt.

Offline checker validates synthetic fixtures only. Trainer separately observes simulator/hardware operation, configuration evidence, diagnosis and safe-change discipline. Use the independent folder instructions and checklist to record acceptance.

Activity12 — Integrated incident and change verification ·Slide712

Use time-correlated service evidence to isolate a trunk fault and verify a narrow change with regression tests and rollback.

Offline checker validates synthetic fixtures only. Trainer separately observes simulator/hardware operation, configuration evidence, diagnosis and safe-change discipline. Use the independent folder instructions and checklist to record acceptance.

Announced v2.0 Transition and Final Revision

Slides 625-700. Effective 3 February 2027; use current v1.1 weighting until 2 February 2027. Include new operational objectives without mislabelling existing EUI-64, L3 LACP or virtualization as new-only.

External practice exam at end: <https://exams.tertiaryinfotech.com/practice-exams/cisco/cisco-ccna>. Practice-site threshold is separate from Cisco certification and WSQ competency criteria.

Assessment Administration

Digital attendance -> assessment attendance -> WA then PP -> LMS submission -> signed Assessment Summary Record. TRAQOM is a separate quality/outcomes feedback survey. Keep answer keys private. Do not substitute activity fixture output for assessed performance.